1. Who was Aryabhata? Briefly explain his contributions to mathematics.

2. Describe the structure of the Aryabhatiya and explain its four chapters.

3. What is meant by the term "Siddhanta"? Who wrote the Pañcasiddhāntikā? Name the five Siddhantas.

4. What is the Kuttaka (Pulverizer) method developed by Aryabhata? Explain its purpose.

5. Explain Aryabhata's rule for π value with up to four decimal accuracy.

6. Explain polygon doubling formula for computing π.

7. Describe Aryabhata's recursive difference sine difference formula.

8. Discuss Aryabhata's contribution to trigonometry and sine tables.

9. Calculate the cube root of the following non-perfect numbers using Vedic mathematics:

29, 38, 53, 59, 67, 79, 107, 131

10. Calculate the cube root of the following perfect numbers using Vedic mathematics:

9261, 74088, 6859, 50653, 389017, 42875, -941192, -226981, 438.976, - 531.441

11. Find out the square root of the following numbers using Vedic mathematics:

2376.592, 15876, 125316, 3.58021, 0.8931, 372.4318

12. Using Kuttaka method (Euclidean Algorithm) find out a solution to the Diophantine equation:

$$47x + 30y = 1$$

13. For each of the following linear Diophantine equations, either find the form of a general solution, or show there are no integer solutions

$a.\ 21x + 14y = 147$ $b.\ 30x + 47y = -11$ $c.\ 21x + 14y = 146$

$d.\ 30x + 47y = 2$ $e.\ 4x - 6y = 77$ $f.\ 4x - 6y = 78$

14. Ekanyunena Purvena (By one less than the previous one) : i. 9999 X 2378, ii) 999 X 324567

15. Anurupyena (Proportionately): $a.\ 46 \times 44$ $b.\ 48 \times 42$ $c.\ 61 \times 43$

16. Compute the square of the following integers:

i. 149, ii. 642, iii. 1024, iv. 1777, v. 9614

17. Find the positive square roots for each of the followings:

ii. 390625, ii. 1234321, iii. 285156, iv. 31329, v. 135424

18. Write the name of the following numbers mentioned in the Vedic literatures: 10^7, 10^9, 10^{11}, 10^{12}, 10^{14}, 10^{16}, 10^{17}, and 10^{53}